

CSE446: Blockchain & Cryptocurrencies

Lecture – 6: Money and Cryptocurrency



Inspiring Excellence

Agenda

- Cryptocurrency
- Distributed Ledger/Blockchain concept
- Bitcoin

Crypto~currency

- Based on cryptography
- Derives trust from mathematical properties/algorithms/systems
- Based on established, trusted cryptographic primitives NOT from chemical/physical properties
- NOT from coercive (imposed) Legal Tender statutes
 - Legal tender is a medium of payment recognised by a legal system to be valid for meeting a financial obligation

Fiat money vs Crypto~currency

- Holder has ownership
 - Like any Fiat money, a crypto-currency is a bearer instrument
 - But provides better security as it is important to prove ownership
- No other records kept as to identify an owner
- Easy to keep anonymous
- Hard or impossible to replace if lost or stolen



https://upload.wikimedia.org/wikipedia/en/9/94/1000_Bangladeshi_taka_Obs_2011.jpg

Bitcoin

- Decentralised, Distributed, Voluntary
- No central issuing or verification authority, no “Bitcoin Corp”
 - Bitcoin Foundation (bitcoinfoundation.org)
 - Growing numbers of entrepreneurs accepting or basing new business concepts on Bitcoin
- Relative to other bearer instruments (currency or fiat currency)
 - Easier to transport anywhere in the world
 - Easier to secure, even provides better security
- Relative to other electronic currencies
 - Immune to sovereign censorship, shutdown, or confiscation
 - Immune to inflation and bank defaults

Bitcoin challenges

- Creation of a virtual coin/note
 - How is it created in the first place?
 - How do you prevent inflation? (What prevents anyone from creating lots of coins?)
- Validation
 - Is the coin legit?
 - How do you prevent a coin from double-spending?
- Trust on third-parties
 - Rely on proof instead of trust
 - Verifiable by everyone
 - No central bank or clearing house

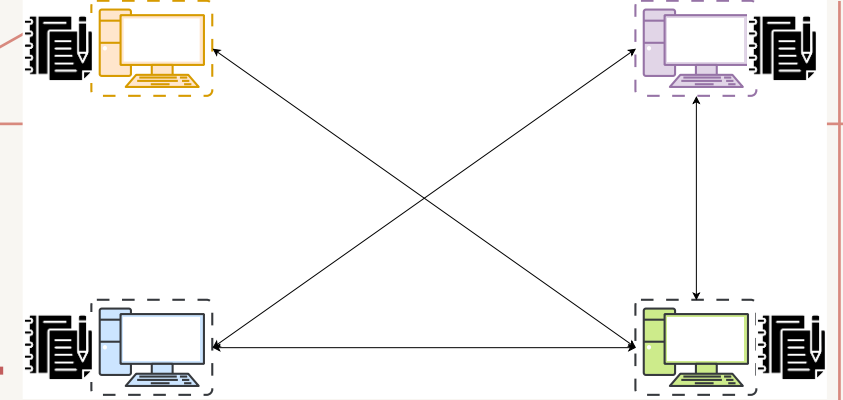
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Blockchain/Distributed Ledger is the solution

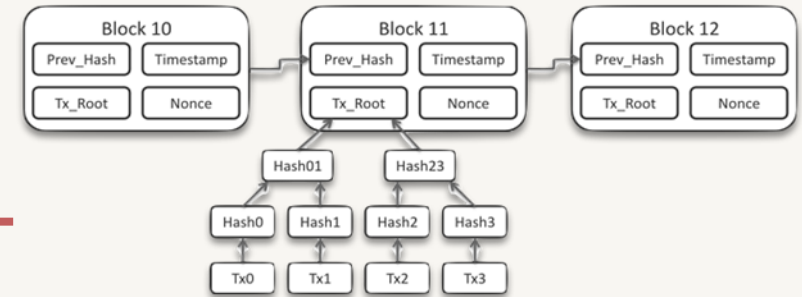
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Distributed Ledger



- A general ledger is the heart of any banking and financial institutions
- To tackle the centralised trust issues
 - dissolve the centralised trust, replace it with a decentralised trust
- One way: distribute the ledger over as many entities as possible
 - Hence the notion of distributed ledger
- Slight difference in meanings between blockchain and distributed ledger
 - A blockchain is just an example of a distributed ledger

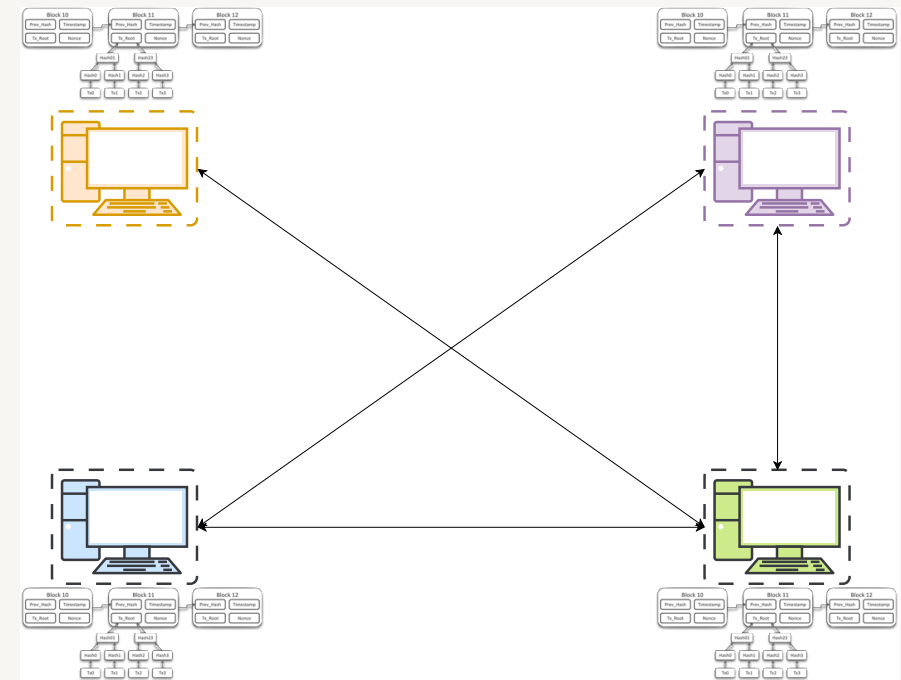
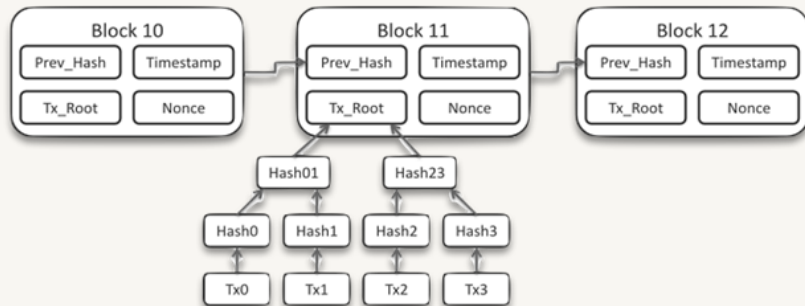
Blockchain



- The blockchain is a distributed record of transactions structured in a specific way
- These transactions are grouped together following specific sets of rules
 - These groups are known as Blocks
- Blocks are linked together with specific rules, thus forming the chain
- Blockchain is a chain of blocks, where each block maintains a specific data structure

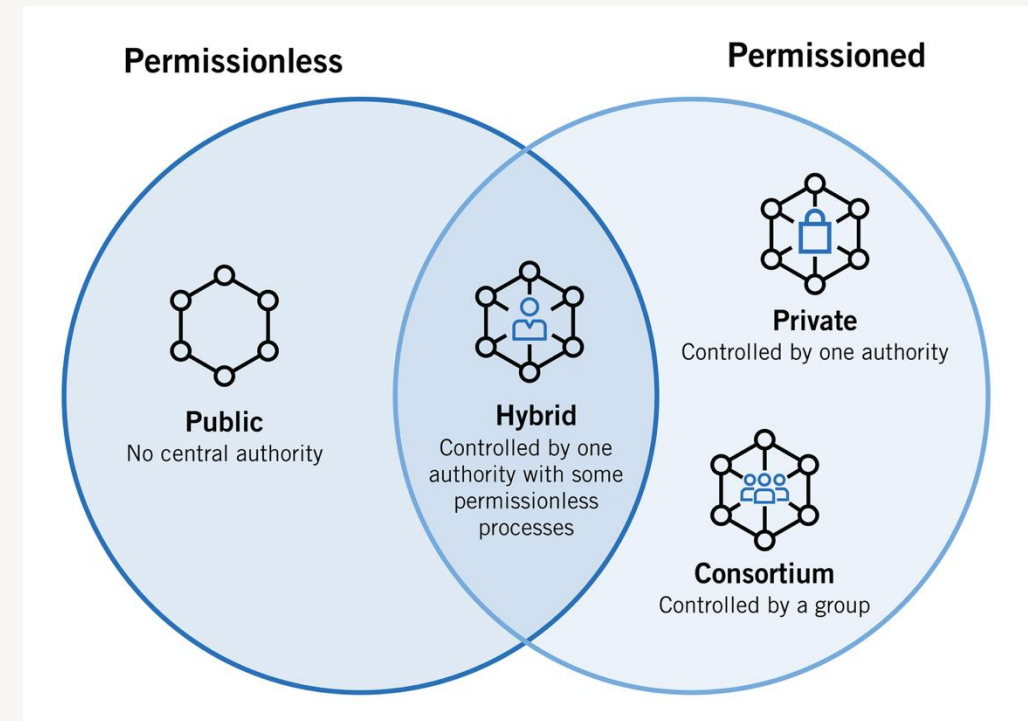
Blockchain

Even though a blockchain is just a data structure, however, it implies a distributed data structure



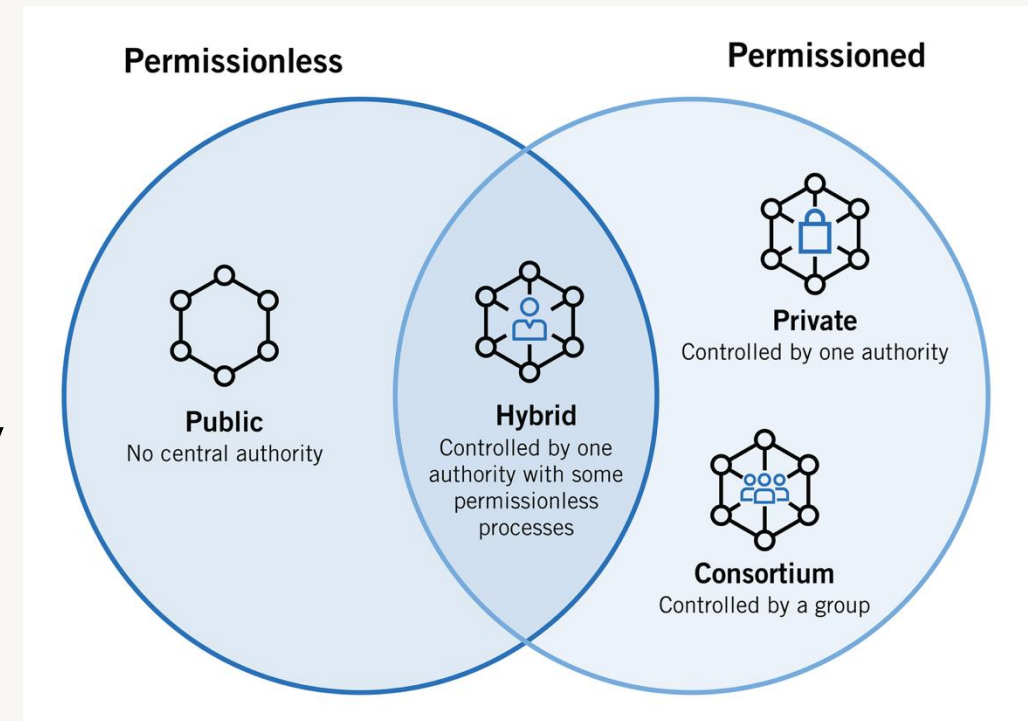
Blockchain types

- Depending who can access (read) or write data from/to a blockchain, there could be four types of blockchain
 - Public (permissionless) blockchain
 - Private (permissioned) blockchain
 - Hybrid blockchain
 - Consortium blockchain



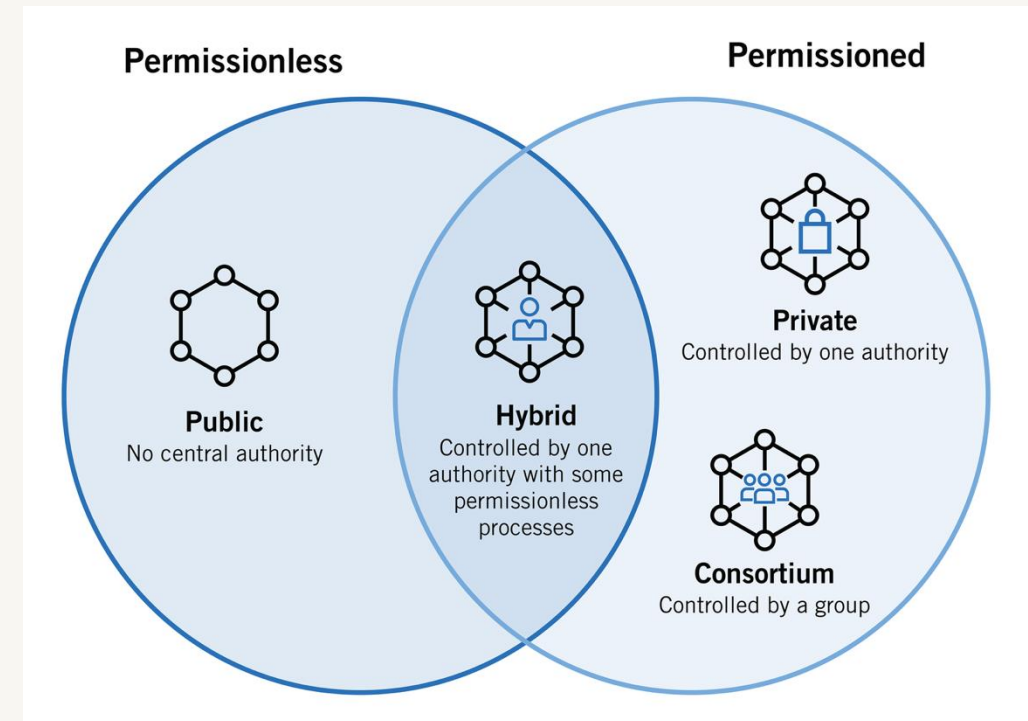
Public blockchain

- A public blockchain is an open blockchain which allows everyone to join in the network
- Everyone can write into the blockchain following specific rules
- All data can be read by all
- Everyone can verify all data in the blockchain
- No one is trusted & there is no central authority
- Almost all crypto-currency blockchains are public, e.g. Bitcoin, Ethereum, Solana, Cardano



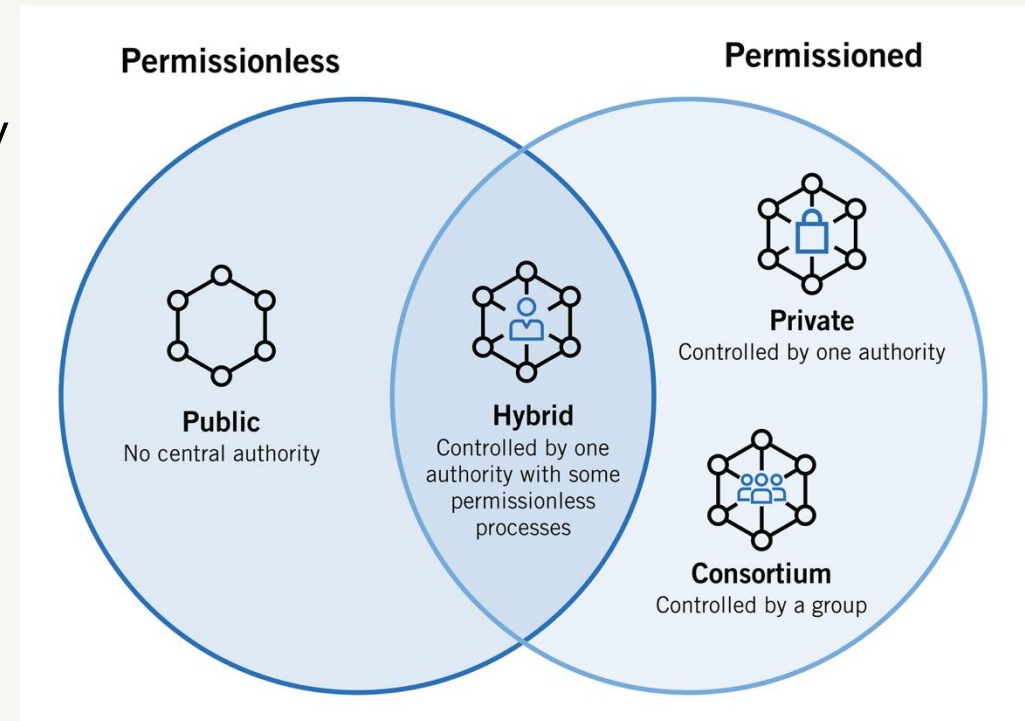
Private blockchain

- A private blockchain is controlled by a single authority (e.g. a bank or an org)
- The network is private and set up between trusted partners (e.g. different bank branches)
- Sets own rules and regulations
- Restricted read/write access so that only authorised parties can participate
- Examples: Hyperledger Fabric, Hyperledger Sawtooth, Corda, etc.



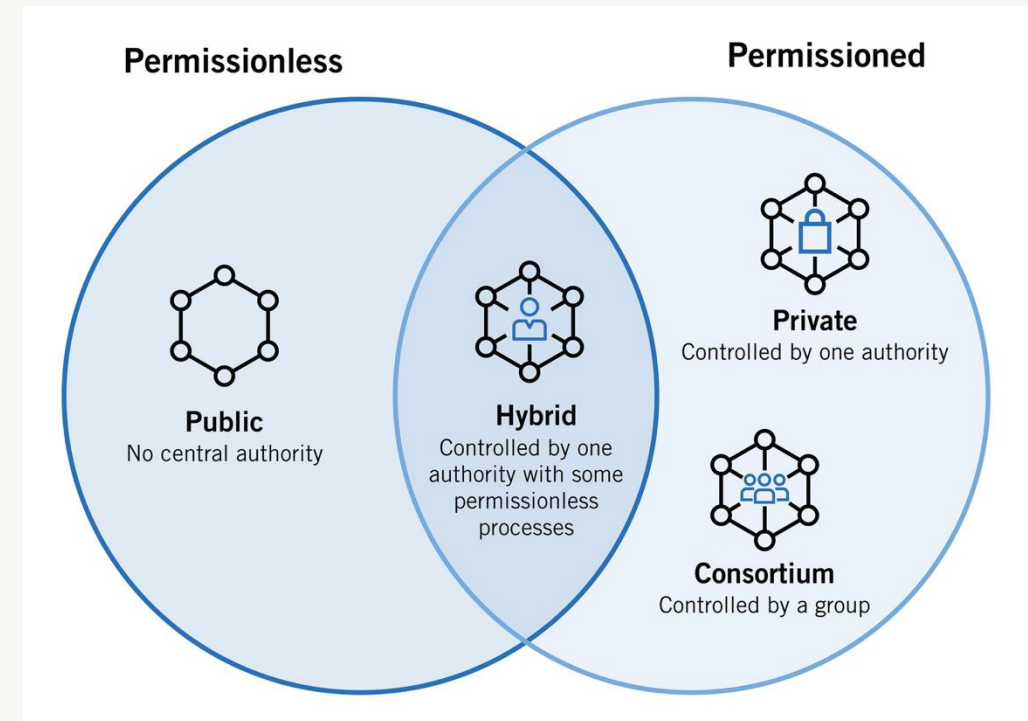
Hybrid blockchain

- A hybrid blockchain is a combination of public and private blockchain
- It is usually controlled by a central authority
- The authority sets up the rules
- Everyone can read data from the blockchain
- Write access is restricted
- Examples: LTO Network, Sovrin



Consortium blockchain

- It is a private blockchain controlled by a set of private entities (e.g. a consortium of banks in Bangladesh)
- They set up their own rules and regulations
- Read and write access are controlled and are only allowed to the authorised entities
- Can be set up with private blockchains



Summary

Types	Read	Write
Public	All	All
Private	Restricted	Restricted
Hybrid	All	Restricted
Consortium	Restricted	Restricted

